3.37 Jacobson Rings

Definition 3.37.1 (Jacobson Radical)
*Let $\mathfrak{a} \triangleleft A$ be an ideal. Define the **Jacobson Radical** of $\mathfrak{a}$ to be*

$$\sqrt{\mathfrak{a}}^J := \bigcap_{\mathfrak{a} \subseteq \mathfrak{m}} \mathfrak{m}$$

Note by (3.3.45) and (3.3.59)

$$\sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{a}}^J$$

Proposition 3.37.2 (Jacobson Ring)
Let A be a ring the following are equivalent

a) For any ideal $\mathfrak{a}$, $\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{a}}^J$
b) For any radical ideal $\mathfrak{a} = \sqrt{\mathfrak{a}}^J$
c) For any prime ideal $\mathfrak{p} = \sqrt{\mathfrak{p}}^J$

*We say such a ring is a **Jacobson ring**.*

Proof. a) $\implies$ b) This is clear because in this case $\mathfrak{a} = \sqrt{\mathfrak{a}}$.

b) $\implies$ c) This is clear because a prime ideal is radical.

c) $\implies$ a) By (3.3.45)

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{p}} \mathfrak{p} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{p} \subseteq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{m}} \mathfrak{m}$$

as required ∎

We prove the Weak Nullstellensatz implies the following result

Proposition 3.37.3 (Strong Nullstellensatz)
Let A be a finitely-generated k-algebra. Then it is a Jacobson ring.

Proof. **TODO** ∎